

Yeonsu Chang

Integrated Ph.D Candidate

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Graph Theory

Structural Graph Theory

Parameterized Complexity

Reduced Parameter

Research Statement

My research focuses on structural aspects of graphs and related width parameters, with emphasis on characterizations and algorithmic consequences in parameterized complexity.

Education

Integrated Ph.D in Mathematics

Mar 2022 – Present

Hanyang University, Seoul, South Korea

Advisor: O-joung Kwon

B.S. in Mathematics

Mar 2015 – Feb 2022

Hanyang University, Seoul, South Korea (incl. two years of military service)

Military Service

Republic of Korea Army

Jan 2017 – Oct 2018

Completed mandatory military service (Sergeant)

Journal Papers

3. Shinwoo An, Yeonsu Chang, Kyungjin Cho, O-joung Kwon, Myounghwan Lee, Eunjin Oh, and Hyeonjun Shin. **Pre-assignment problem for unique minimum vertex cover on bounded clique-width graphs**, *Theoretical Computer Science* 1084 (Sep, 2026), 116173, *AAAI2025 accepted*, [arXiv:2408.09591](https://arxiv.org/abs/2408.09591)
2. Yeonsu Chang, O-joung Kwon, and Myounghwan Lee. **A new width parameter of graphs based on edge cuts: α -edge-crossing width**, *Discrete Applied Mathematics* 380 (Feb, 2026), *WG23 accepted*, [arXiv:2302.04624](https://arxiv.org/abs/2302.04624)
1. Yeonsu Chang, Sejin Ko, O-joung Kwon, and Myounghwan Lee. **A characterization of graphs of radius- r flip-width at most 2**, *Discrete Mathematics* 348(4) (Apr, 2025), 114366 [arXiv:2306.15206](https://arxiv.org/abs/2306.15206)

Refereed Conference Papers without Journal Version

1. Édouard Bonnet, Yeonsu Chang, Julien Duron, Colin Geniet, and O-joung Kwon. **Moderately beyond clique-width: reduced component max-leaf and related parameters**, *ESA 2026 accepted*, [arXiv:2604.19138](https://arxiv.org/abs/2604.19138)

Preprints

1. Yeonsu Chang, Shenwei Huang, and Yidong Zhou. **The optimal chromatic bound for even-hole-free graphs without induced seven-vertex paths**, [arXiv:2602.04403](https://arxiv.org/abs/2602.04403)

Talks

Korean Mathematical Society 80th Anniversary Conference <i>The optimal chromatic bound for (even-hole, P_7)-free graphs</i>	Jun 22–25, 2026
Korean Student Combinatorics Workshop 2026 Winter (KSCW2026W) <i>Structural and Algorithmic properties of Reduced Component Max-Leaf</i>	Feb 2–6, 2026
2025 Korean Mathematical Society Annual Meeting <i>Structural and Algorithmic properties of Reduced Component Max-Leaf</i>	Oct 22–24, 2025
Math Colloquium at Hanyang University <i>Combinatorial game and graph structure: cops and robber game and flipper game</i>	Nov 4, 2024
2023 KMS Annual Meeting <i>A characterization of graphs of radius-r flip-width at most 2</i>	Oct 26–28, 2023

Teaching Experiences

Linear Algebra 1, Hanyang University	Spring 2025
Math Capstone PBL (Solving Graph Problems), Hanyang University	Fall 2024
Linear Algebra 1, Differential Geometry, Hanyang University	Spring 2024
Head TA, Hanyang University	Fall 2022 – Spring 2024
Math Capstone PBL (Applications of graph algorithms), Hanyang University	Fall 2022
Linear Algebra 1, Hanyang University	Spring 2022

Recent Directions

- Reduced parameter
- Reduced component max-leaf
- Chiboundedness
- Characterization of graph classes
- Sparsity